LAGOS BUSINESS SCHOOL · MEMBA 14 · EEGB · PART I

SESSION 5

Market Failures & Government Intervention



From markets, to macro, to decisions that matter — in Nigeria and the world.

Reading: Externalities: Justification for Public Goods? (210-023-1) · A Framework to Think About Pollution (UVA-GEM-0104) · Facilitator: Kelechi Nnamdi

RECAP

Session 4 in Three Ideas



Market structure

Fewer, bigger rivals mean more power to raise prices — and deadweight loss.



Platforms

Network effects, switching costs and data can tip a market towards one winner.



Competition policy

Define the market, measure concentration, judge conduct — not size.

TODAY'S PLAN

Session 5 at a Glance — 75 Minutes

When markets get it wrong, when governments get it wrong — and the petrol subsidy as a case study in both.



?

Recap & Hook

Who really benefited from cheap petrol?

⚠

Four Market Failures

Externalities, public goods, information, market power

⚙

The Policy Toolkit

Taxes, cap-and-trade, rules, nudges, market design

26 FD

Subsidy Case & Lab

Incidence, removal, compensation

BY THE END OF TODAY

Session 5 Learning Objectives

You will be able to:

01

Diagnose a Market Failure

Recognise externalities, public goods, asymmetric information and market power in your own industry.

02

Choose the Right Fix

Compare Pigouvian taxes, cap-and-trade, regulation, nudges and market design — and when each works.

03

Analyse a Subsidy

Work out who really gains from a subsidy, who loses when it goes, and what compensation should look like.

THE HOOK

Who Really Got Cheap Petrol?



“The petrol subsidy was the one benefit ordinary Nigerians got from the oil wealth.”

A popular claim. The economist's questions:

Who buys the most petrol?

Households with cars and big generators — mostly the better-off — use many times more than the poor.

Where did it leak?

Cheap petrol was smuggled to neighbouring countries where pump prices were much higher.

What else could the money buy?

Every naira spent on the subsidy was a naira not spent on schools, clinics or roads.

NIGERIAN BUSINESS CASE

Petrol Subsidy Removal

The situation

For decades Nigeria held the pump price of petrol below its cost, paying the difference from public funds. In mid-2023 the subsidy was removed and the pump price rose from about **₦185 to ₦617** per litre.

The decision question

Who **gained** from the subsidy, who **loses** from its removal, and what should **replace** it?

The data we have

fuel_subsidy_incidence.csv — household spending, petrol use, food and transport shares, vehicle and generator ownership, by consumption decile. *Stylised teaching data.*

What a good answer looks like

A policy recommendation: how the savings should be used, and how to **protect poor households** at a sensible fiscal cost.

PART

01

When Markets Fail

Four reasons prices give the wrong signal.

Why Free Markets Sometimes Get It Wrong

Externalities

A cost or benefit falls on people **outside** the deal. Generator fumes, gas flaring, traffic congestion — or the benefit of a vaccinated neighbour.

Public goods

Nobody can be excluded, and one person's use does not reduce another's: street lighting, flood defences, national defence. Everyone waits for someone else to pay.

Asymmetric information

One side knows more: the used-car seller, the borrower, the insurance buyer. Good products can be driven out by bad ones.

Market power

Session 4: a dominant firm restricts output and raises prices.

The Generator Next Door

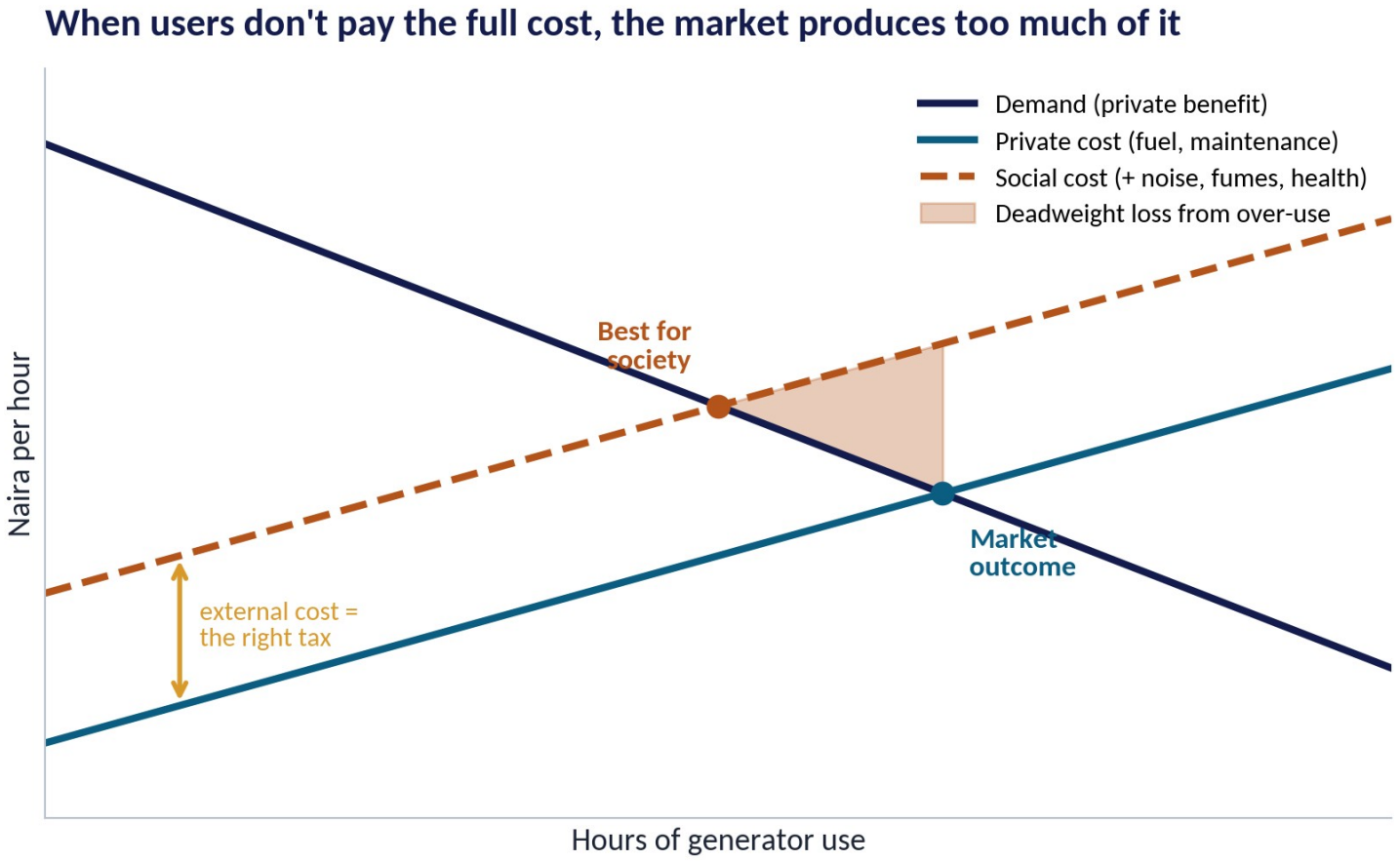


Fig S05-F01 · replicate: s05_figures.R / .py

WHAT TO SEE

- The generator owner pays for fuel and maintenance — the **private cost**.
- Neighbours pay too, in noise, fumes and health — the **external cost**.
- The market uses generators up to where private cost meets benefit: **too much**.
- A tax equal to the external cost moves use to what is **best for society** — the idea behind a **Pigouvian tax**.

The Lemons Problem

IN PLAIN ENGLISH

When buyers cannot tell good from bad, they pay an **average** price. Owners of good products then refuse to sell, the average quality falls, and the market can **unravel**.

EXAMPLE

Lenders cannot tell safe from risky small businesses, so they charge everyone high rates — which drives away the safest borrowers first. Credit bureaus and collateral registries exist to fix this.

THE FORMULA

hidden quality → average price → good sellers leave

WHY A MANAGER CARES

Your business can **profit from solving** information problems: warranties, ratings, certification, track records, and data-based credit scoring all build trust.

Private Goods vs Public Goods

Private goods

Excludable and rival

- You can stop non-payers using it
- My use leaves less for you
- Markets supply them well
- Bread, phones, haircuts

Public goods

Non-excludable and non-rival

- Hard to stop non-payers
- My use does not reduce yours
- Markets under-supply them: free riding
- Street lights, flood control, basic research, security

*Many Nigerian firms **self-supply** public goods — power, water, security, roads to the factory. That is a hidden tax on doing business here.*

PART

02

The Policy Toolkit

Prices, quantities, rules and nudges.

Matching the Tool to the Failure

Tool	How it works	Strength	Weakness
Pigouvian tax	Charge the polluter the external cost per unit	Firms cut pollution where it is cheapest	Hard to set the right rate; unpopular
Cap-and-trade	Fix total pollution; issue tradable permits	Certainty about the total	Price can swing; needs strong monitoring
Regulation	Set standards: emission limits, bans	Simple and visible	Same rule for all, so costly for some
Nudges	Change defaults and framing: auto-enrol pensions	Cheap; keeps choice	Small effects; can be manipulative
Market design	Build the market: auctions, registries, ratings	Fixes information and allocation	Needs institutions and data

*The economist's preference: **price the harm** (tax or permits) and let people decide how to adjust — rather than dictate one solution.*

Pigouvian Taxes vs Cap-and-Trade

IN PLAIN ENGLISH

A **tax** fixes the **price** of pollution and lets the quantity adjust. **Cap-and-trade** fixes the **quantity** and lets the price adjust. Both make polluting costly, so firms that can cut cheaply do the most cutting.

EXAMPLE

A gas-flaring penalty per thousand cubic feet is a tax-style tool; a carbon market with a national emissions cap is the permit approach. Session 13 returns to carbon pricing.

THE FORMULA

tax: price fixed, quantity varies · cap: quantity fixed, price varies

WHY A MANAGER CARES

For your business: either tool turns pollution into a **cost line**. Firms that cut emissions early gain a cost advantage — or permits to sell.

Interventions Can Fail Too

IN PLAIN ENGLISH

Governments face their own problems: **poor information**, **lobbying** by those who benefit, **leakage** and **corruption**, and policies that are easy to start but hard to stop.

EXAMPLE

The petrol subsidy was meant to share oil wealth, but it favoured car and generator owners, encouraged smuggling across borders, and became one of the largest items in the federal budget.

THE FORMULA

market failure ≠ automatic case for intervention

WHY A MANAGER CARES

The right question is not 'market or government?' but 'which **imperfect** arrangement does least harm here?'

Quick Check: What Would You Say?

?

A Lagos estate's residents all run diesel generators, and everyone complains about the noise and fumes. Which policy best fits the economics of externalities?

- **A** Ban generators outright
- **B** Charge a fee per generator-hour that reflects the harm, and use the money to co-fund a shared, cleaner power supply
- **C** Do nothing — residents chose to live there
- **D** Subsidise diesel so generators are cheaper to run

Discuss with your neighbour for 60 seconds — answer and reasoning in the speaker notes.

PART

03

The Petrol Subsidy

Who gained, who pays, and what next.

Who Really Benefits From a Subsidy?

IN PLAIN ENGLISH

A **per-litre** subsidy goes to whoever **buys the most litres**. Rich households use far more petrol — for cars and generators — so they receive far more in naira, even if the subsidy is a bigger **share** of a poor family's budget.

EXAMPLE

Before removal (stylised): ₦315 of subsidy per litre. A household in the richest 10% using 150 litres a month got **₦47,250**; one in the poorest 10% using 4 litres got **₦1,260**.

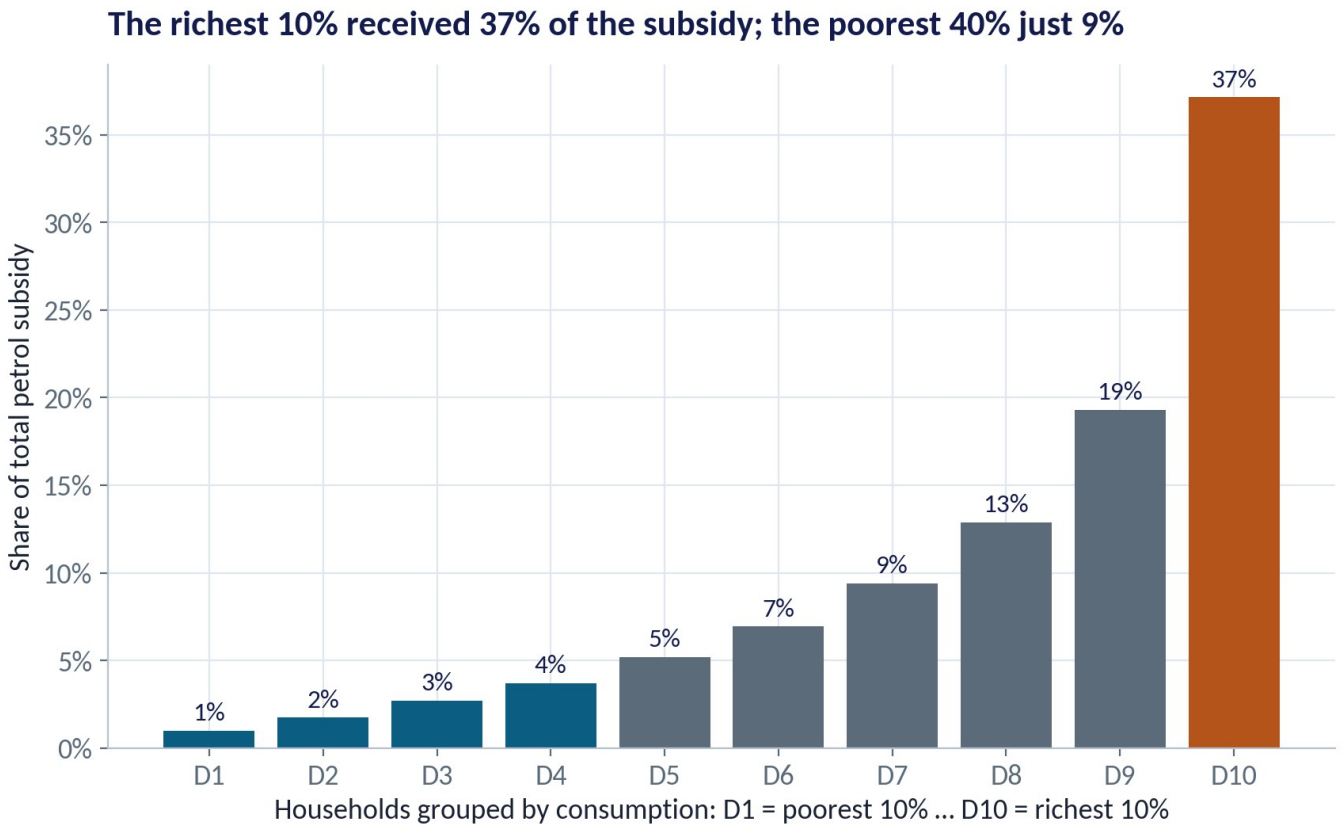
THE FORMULA

$$\text{benefit} = \text{litres used} \times \text{subsidy per litre}$$

WHY A MANAGER CARES

Universal price subsidies are an expensive way to help the poor. Most of the money goes to people who do not need it.

Most of the Money Went to the Better-Off



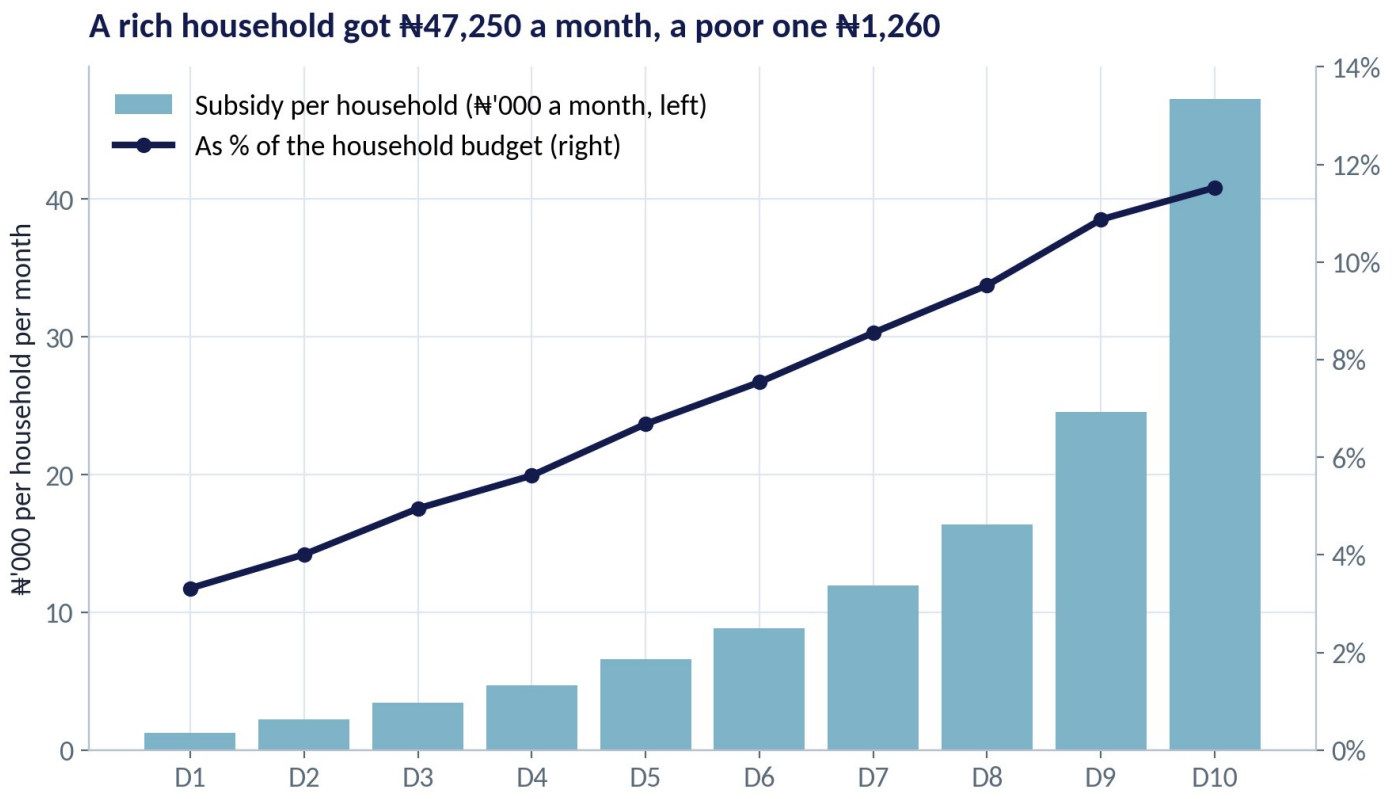
Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S05-F02 · replicate: s05_figures.R / .py

WHAT TO SEE

- The richest 10% of households received about **37%** of the household petrol subsidy.
- The poorest 40% received only about **9%**.
- In naira, the richest households got about **38 times** as much as the poorest.
- This pattern — subsidies captured by the better-off — is found in fuel subsidies across many countries.

Naira Received vs Share of Budget



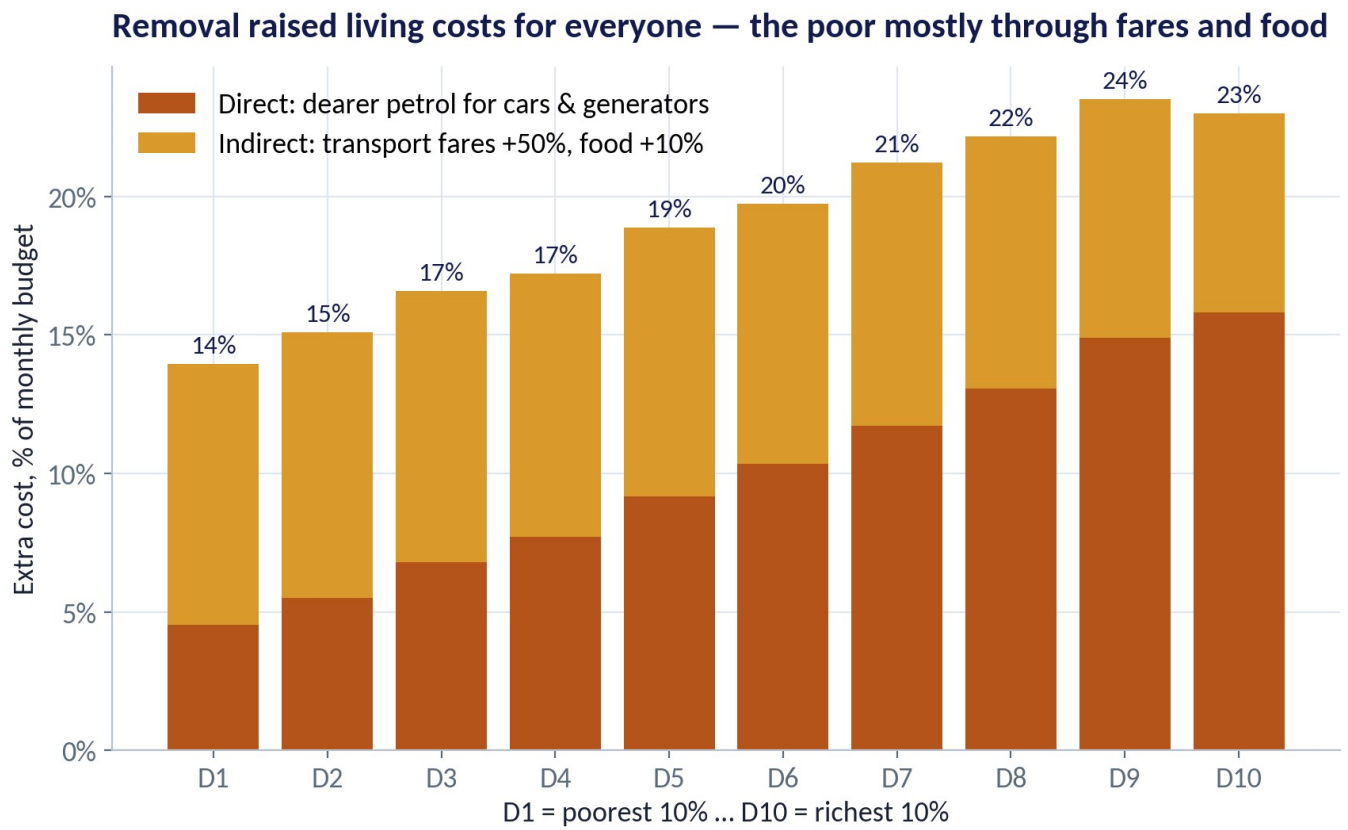
Source: EEGB stylised teaching data (data/). Not official statistics.

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WHAT TO SEE

- Bars: the subsidy in **naira** rose steeply with income.
- Line: as a **% of the budget**, it also rose — from about **3%** for the poorest to **12%** for the richest.
- Poor households mostly use petrol **indirectly**: through bus fares and the cost of moving food.
- That indirect channel is why removal still hurt the poor badly.

The Cost of Removal



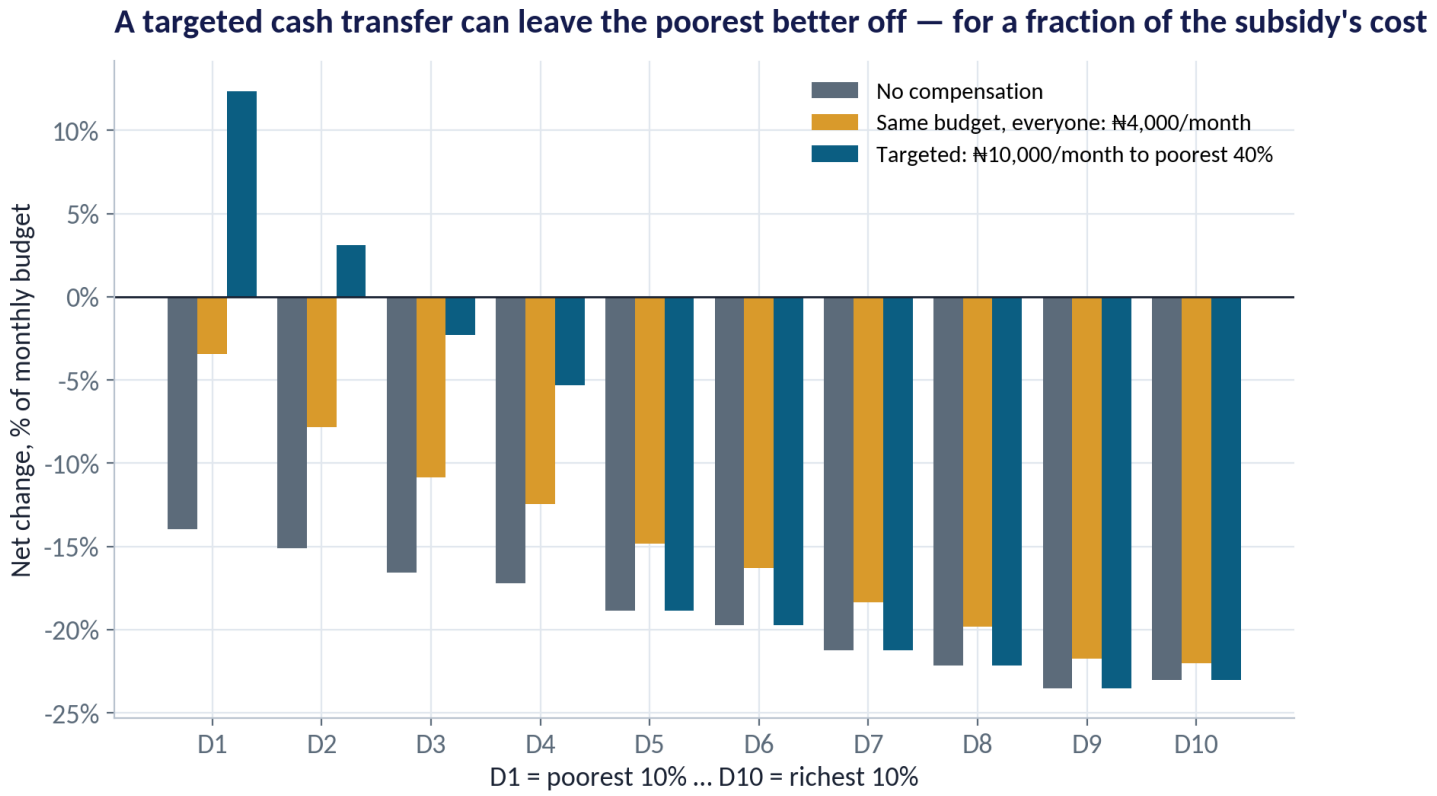
Pump price ₦185 → ~~₦617~~ per litre; indirect effects are stylised assumptions. Source: EEGB stylised teaching data.

Fig S05-F04 · replicate: s05_figures.R / .py

WHAT TO SEE

- Direct cost: petrol rose from **~~₦185~~ to ~~₦617~~** a litre.
- Indirect cost (our assumptions): bus fares **+50%**, food **+10%**.
- Extra cost: about **14%** of the budget for the poorest decile, **23%** for the richest.
- For the poor, most of the hit is **indirect** — and they have the least room to cut back.

Targeted vs Universal Support



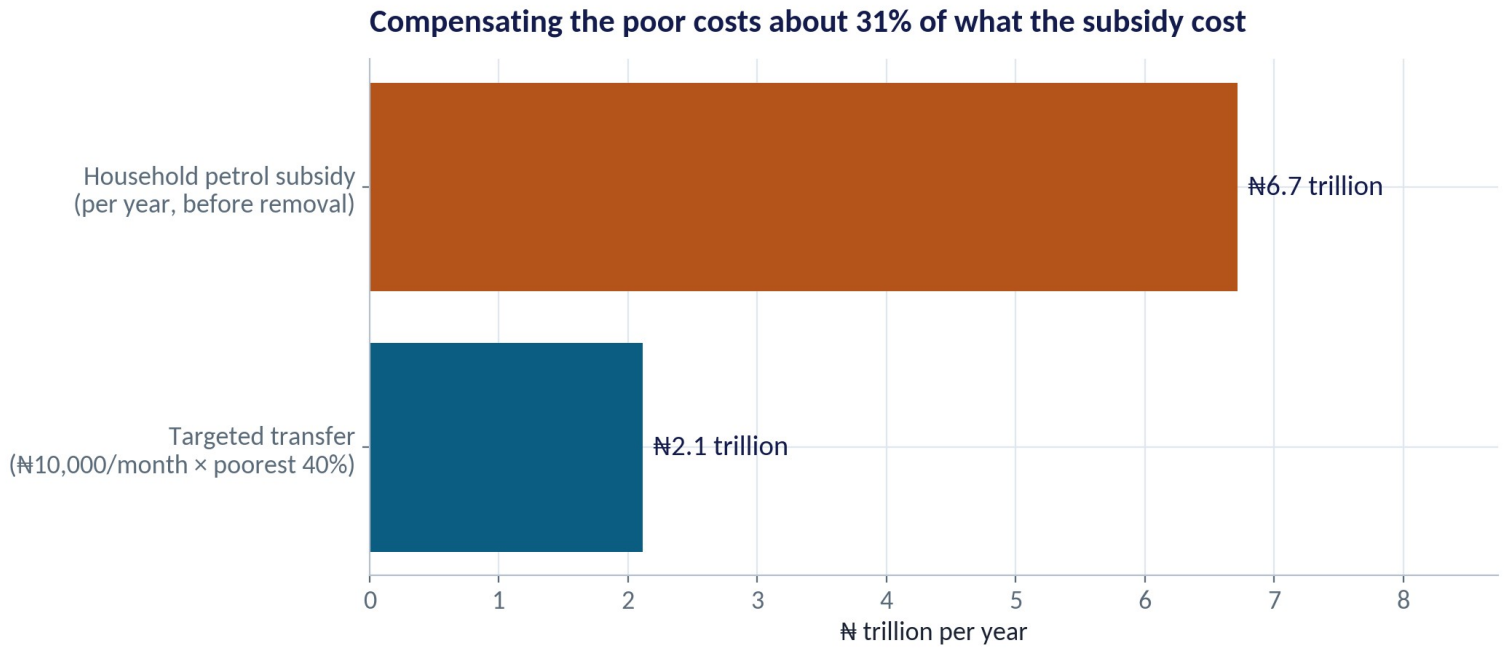
Source: EEGB stylised teaching data (data/). Not official statistics.

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WHAT TO SEE

- A **₦10,000 a month** transfer to the poorest 40% leaves the poorest two deciles **better off** than before removal.
- The same budget spread over everyone gives just **₦4,000** each — too little to protect the poor.
- Better-off households still bear the full cost — but they also gained most from the subsidy.
- Targeting needs a reliable **social register**, digital IDs and payment channels.

What the Savings Can Buy



Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S05-F06 · replicate: s05_figures.R / .py

WHAT TO SEE

- The household petrol subsidy cost about **₦6.7 trillion** a year (stylised).
- The targeted transfer costs about **₦2.1 trillion** — roughly **31%** of that.
- The rest can fund transport (e.g. CNG buses), health and education — or cut borrowing.
- Credibility matters: people accept reform more readily when they can **see** where the savings go.

WHERE MANAGERS GO WRONG

Common Mistakes in Thinking About Intervention

THE TRAP	THE BETTER HABIT
✗ “A price subsidy is the best way to help the poor.”	✓ Check **who consumes** the subsidised good. Targeted cash usually reaches the poor more cheaply.
✗ “Removing the subsidy only hurts car owners.”	✓ Count **indirect effects**: fares, food, and the cost of running small businesses.
✗ Banning the polluting activity outright.	✓ **Price** the harm and let people adjust; ban only where harm is severe.
✗ Assuming any market failure justifies intervention.	✓ Compare against **government failure**: leakage, lobbying, weak capacity.
✗ Ignoring the business opportunity.	✓ Firms that **solve** failures — ratings, clean power, logistics — create value.

CONCEPT CHECK

Quick Check: What Would You Say?

?

Our data show the richest 10% got 37% of the subsidy but face the largest % cost from removal (23% of budget vs 14% for the poorest). What follows?

A. Removal was regressive because the rich lost more

B. The rich lose more in % terms, but they gained most from the subsidy and can absorb it; protecting the poorest — who have little margin — is the priority for compensation

C. Compensation should go mainly to the richest decile

D. The subsidy should be restored

Discuss with your neighbour for 60 seconds — answer and reasoning in the speaker notes.

HANDS-ON LAB

Lab: Subsidy Incidence and Compensation

YOU WILL NEED

Data
`data/fuel_subsidy_incidence.csv`

R script
`sessions/s05/R/s05_lab.R`

Python script
`sessions/s05/python/s05_lab.py`

⏱ **20 minutes**

1. Compute the subsidy each decile received, in **naira** and as a **% of budget**.
2. Find the **shares** of the richest 10% and the poorest 40%.
3. Estimate the **direct and indirect** cost of removal; change the fare and food assumptions.
4. Cost a **targeted cash transfer** and compare it with the old subsidy bill.
5. Find the transfer that leaves deciles 3 and 4 **no worse off**.

Deliverable: a two-sentence recommendation on how the savings should be used.

WORDS TO OWN

Today's Vocabulary in Plain English

Market failure	When a free market produces the wrong amount of something.
Externality	A cost or benefit that falls on people outside the transaction.
Public good	Non-excludable and non-rival, so markets under-supply it.
Free riding	Enjoying a benefit without paying towards it.
Asymmetric information	One side of a deal knows more than the other.
Pigouvian tax	A tax equal to the external cost of an activity.
Cap-and-trade	A fixed total of pollution permits that firms can trade.
Nudge	A change in defaults or framing that steers choices.
Incidence	Who actually bears a tax or receives a subsidy.
Government failure	When intervention makes outcomes worse.

KEY TAKEAWAYS

Session 5 in 5 Ideas

01 Markets fail through **externalities, public goods, information gaps and market power**.

02 Fix them by **pricing the harm** (taxes, permits), rules, nudges or **market design**.

03 Governments fail too: compare **imperfect alternatives**.

04 A per-litre subsidy goes to **whoever uses the most** — usually the better-off.

05 Protect the poor with **targeted transfers**, and show where the savings go.



NEXT › Session 6: AD–AS & Inflation

Before next session: Read the chapters on aggregate demand and aggregate supply in Blanchard, Macroeconomics. Bring one price your company raised in 2024 and say whether the cause was demand (customers buying more) or supply (your costs rising).